Advanced Quantum Mechanics

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Chapter 1

Advanced Quantum Mechanics I: Quantum Dynamics and Symmetries

This chapter rebuilds the basic mathematical spine of the lecture: quantum states as vectors, particles in the position representation, the Hamiltonian-generated unitary time flow, and the symmetry principle that identifies conserved quantities by commuting operators. The goal is to move from abstract postulates to the concrete rules used in practical calculations, with a consistent convention and clear links between infinitesimal generators and finite transformations.

1.1 Review of the quantum kinematic structure

A physical system is represented by vectors in a (complex) vector space. A state is written as a ket $|\psi\rangle$, and its dual by $\langle\psi|$. The inner product $\langle\phi|\psi\rangle$ is a complex number associated to each pair of states.

$$\langle\psi|\psi\rangle = 1 \quad \text{for a normalized state.} \tag{1.1}$$

An observable A is represented by a Hermitian operator $\hat{A}$ ($\hat{A}^\dagger = \hat{A}$). The eigenvalue problem has physical meaning:

$$\hat{A}\,|a\rangle = a\,|a\rangle\,, \qquad a \in \mathbb{R}. \tag{1.2}$$

If the system is in $|a\rangle$, measurement of A is certain to give a. Operational distinguishability is encoded by orthogonality:

$$|\phi\rangle \text{ and } |\psi\rangle \text{ are distinguishable with certainty } \iff \langle\phi|\psi\rangle = 0. \tag{1.3}$$

In finite dimensions these are ordinary vectors; for particles and fields, the basis index becomes continuous and states are represented by functions.

1.2 Particle kinematics in the position representation

For a one-dimensional particle, define position eigenstates $|x\rangle$ by

$$\langle x|x'\rangle = \delta(x - x'). \tag{1.4}$$

The corresponding wavefunction is the position amplitude

$$\psi(x) = \langle x|\psi\rangle. \tag{1.5}$$

Its modulus squared is the position probability density:

$$p(x) = |\psi(x)|^2 = \psi^*(x)\psi(x), \tag{1.6}$$

and for normalizable states $\int dx\, |\psi(x)|^2 = 1$.

The position operator acts by multiplication:

$$(\hat{x}\psi)(x) = x\,\psi(x). \tag{1.7}$$

Distributional caveat. The "vectors" $|x\rangle$ are not ordinary square-integrable functions. Their coordinate overlap is the Dirac delta, and the position eigenfunctions are idealized delta spikes rather than physical packet states. Physical states are square-integrable approximants (e.g., narrow Gaussians).

1.3 Momentum in the same representation

The momentum observable is represented by a differential operator:

$$(\hat{p}\psi)(x) = -i\hbar\,\frac{d\psi(x)}{dx}. \tag{1.8}$$

(If $\hbar = 1$, drop it from formulas.)

Momentum eigenstates satisfy

$$\hat{p}\phi_p(x) = p\,\phi_p(x). \tag{1.9}$$

A convenient representative is

$$\phi_p(x) = \frac{1}{\sqrt{2\pi\hbar}}\,e^{ipx/\hbar}, \tag{1.10}$$

which has $|\phi_p(x)|^2 =$ constant: momentum certainty implies complete delocalization in position. This is the uncertainty tradeoff in precise mathematical form.

1.4 Time evolution as a unitary transformation

Time evolution is represented by a family $U(t)$ of operators mapping each state to a later state:

$$|\psi(t)\rangle = U(t)\,|\psi(0)\rangle\,, \qquad U(0) = \mathbf{1}. \tag{1.11}$$

A symmetry of information requires inner products to be preserved:

$$\langle\phi|\psi\rangle = \langle U(t)\phi|U(t)\psi\rangle \implies U^\dagger(t)U(t) = \mathbf{1}. \tag{1.12}$$

So $U(t)$ is unitary.

Assuming continuity, for small ϵ,

$$U(\epsilon) = \mathbf{1} - i\epsilon H + \mathcal{O}(\epsilon^2). \tag{1.13}$$

Unitarity to first order gives

$$H^\dagger = H, \tag{1.14}$$

so H is Hermitian and identified as the Hamiltonian.

Now use

$$|\psi(t+\epsilon)\rangle = U(\epsilon)\,|\psi(t)\rangle \tag{1.15}$$

and rearrange:

$$\frac{|\psi(t+\epsilon)\rangle - |\psi(t)\rangle}{\epsilon} = -iH\,|\psi(t)\rangle + \mathcal{O}(\epsilon). \tag{1.16}$$

Hence

$$i\,\frac{d}{dt}\,|\psi(t)\rangle = H\,|\psi(t)\rangle\,. \tag{1.17}$$

For wavefunctions this is

$$i\hbar\,\frac{\partial\psi(x,t)}{\partial t} = H\psi(x,t). \tag{1.18}$$

Finite time is obtained by exponentiation:

$$U(t) = e^{-iHt/\hbar}. \tag{1.19}$$

Stationary states are eigenstates of H:

$$H\,|E_n\rangle = E_n\,|E_n\rangle\,. \tag{1.20}$$

Then

$$|E_n(t)\rangle = e^{-iE_n t/\hbar}\,|E_n\rangle\,, \tag{1.21}$$

so the same Hamiltonian controls both time dependence and allowed energy levels.

1.5 Symmetry as commuting operators

A symmetry V is taken to be a unitary map on states that preserves all physical distinctions. If it represents a true symmetry of dynamics, then evolving then transforming is equivalent to transforming then evolving:

$$V\,U(t) = U(t)\,V, \qquad \forall t. \tag{1.22}$$

Using $U(t) = e^{-iHt/\hbar}$, this is equivalent to

$$[V, H] = 0. \tag{1.23}$$

For a continuous family $V(\epsilon) = \mathbf{1} - i\epsilon G + \mathcal{O}(\epsilon^2)$, the same logic gives

$$[G, H] = 0. \tag{1.24}$$

When $G = G^\dagger$, this is exactly a conserved generator:

$$\frac{d}{dt}\langle G\rangle = \frac{i}{\hbar}\langle [H, G]\rangle = 0. \tag{1.25}$$

Thus symmetry $\leftrightarrow$ conserved quantity is the operator-level statement.

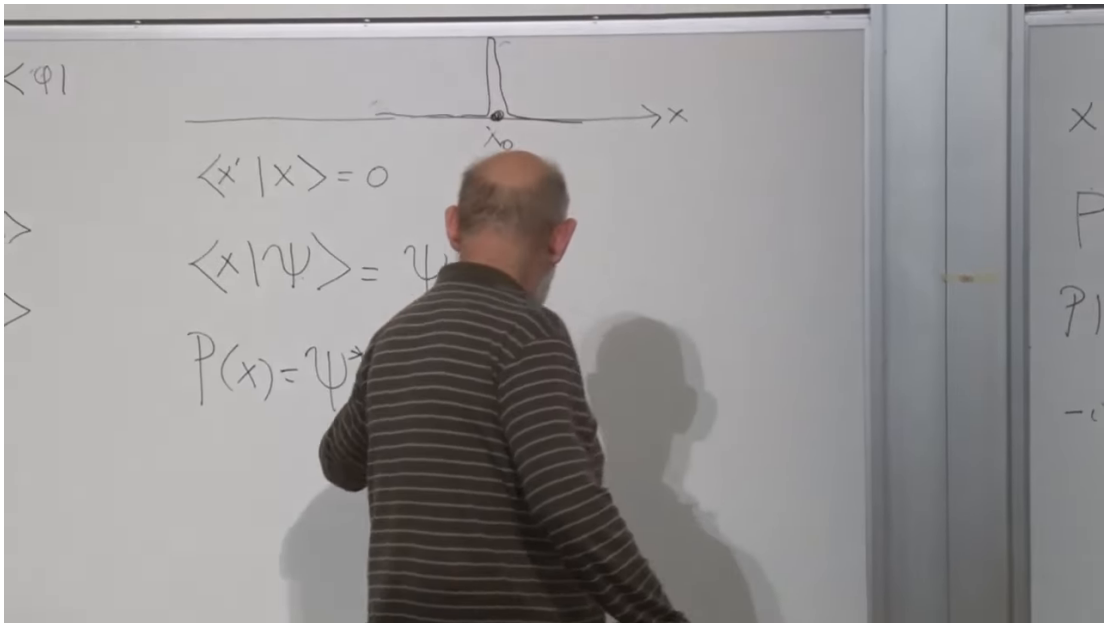


Figure 1.1: Infinitesimal-to-finite evolution structure for $U(\epsilon)$.

$$
\begin{tikzcd}
|\psi_1\rangle \arrow[r, "U(t)"] \arrow[d, "V"'] & |\psi_2\rangle = U(t)\,|\psi_1\rangle \arrow[d, "V"] \\
V\,|\psi_1\rangle \arrow[r, "U(t)"] & V\,|\psi_2\rangle = U(t)\,V\,|\psi_1\rangle
\end{tikzcd}
$$

Figure 1.2: A symmetry operation V commutes with time evolution $U(t)$ if the two routes between states coincide.

1.6 Continuous symmetry example: spatial translations

Define the spatial translation operator $T(\epsilon)$ acting on wavefunctions by

$$(T(\epsilon)\psi)(x) = \psi(x+\epsilon), \tag{1.26}$$

which is a rightward shift by ϵ. For small ϵ,

$$\psi(x+\epsilon) = \psi(x) + \epsilon\,\partial_x\psi(x) + \mathcal{O}(\epsilon^2), \tag{1.27}$$

so

$$T(\epsilon) = \mathbf{1} + \epsilon\,\partial_x + \mathcal{O}(\epsilon^2) = \mathbf{1} - \frac{i\epsilon}{\hbar}\,\hat{p}_x + \mathcal{O}(\epsilon^2). \tag{1.28}$$

Hence

$$T(\epsilon) = \exp\left(-\frac{i}{\hbar}\epsilon\,\hat{p}_x\right), \tag{1.29}$$

so the generator is

$$G_{\text{trans}} = \frac{\hat{p}_x}{\hbar}. \tag{1.30}$$

For a free particle,

$$H = \frac{\hat{p}_x^2}{2m}, \tag{1.31}$$

and $[H, \hat{p}_x] = 0$, so translation is a symmetry and $\hat{p}_x$ is conserved. Thus the same operator appears as both "generator of translations" and "conserved quantity," illustrating the lecture's central symmetry–conservation link.

1.7 Summary

The lecture's core chain is: state vectors → continuous basis amplitudes → operator representation → unitary time flow. Continuity plus unitarity forces $U(t)$ to be generated by a Hermitian Hamiltonian and yields the Schrödinger equations. A symmetry is an additional unitary map compatible with $U(t)$, which becomes a commutator condition with H. For continuous symmetries, infinitesimal generators G define finite transformations via exponentials, and commuting generators are conserved quantities. Translations provide the prototype: $T(\epsilon) = e^{-i\epsilon\hat{p}_x/\hbar}$, and for $H = p^2/2m$, momentum conservation follows directly from $[H, \hat{p}_x] = 0$.

Chapter 2

Symmetry, Degeneracy, and Angular Momentum for Quantum Rotational Systems

This chapter develops a symmetry-first view of rotational quantum systems. We define symmetries as operators commuting with the Hamiltonian, distinguish generic level coincidences from symmetry-protected multiplets, and build angular-momentum structure from the commutator algebra of generators. The particle on a circle provides an explicit worked model where quantized m-values, reflection pairing, and ladder operators emerge from operator algebra rather than ad hoc assumptions.

2.1 Symmetry as an invariance principle

A symmetry is an operation that leaves the quantum description unchanged. For a unitary transformation S, invariance of dynamics is the condition

$$SHS^\dagger = H$$

equivalently

$$[H, S] = 0.$$

Because S commutes with H, Schrödinger evolution is covariant under S: if $|\psi(t)\rangle$ evolves by H, then $S|\psi(t)\rangle$ evolves in the same way. In particular, E-eigenvalues are mapped to the same E-eigenspaces.

Symmetries can be exact (invariance of the full Hamiltonian), approximate (effective in a regime), or accidental (valid only for very specific parameter choices). Our focus is on exact symmetries because they control spectral structure.

2.2 Symmetry examples and the organizing role of operators

The same formal rule $[H, S] = 0$ unifies many familiar examples:

- spatial translations and rotations,

- interchange of identical particles,
- relabeling symmetries of a periodic lattice (discrete translation in crystal index),
- discrete mirror or inversion operations.

In practice, this is less about geometry alone and more about algebraic leverage: a set of commuting operators gives you a good quantum-number basis, while noncommuting families often generate nontrivial multiplets. Symmetry language gives a compact way to classify allowed degeneracies before solving differential equations.

Figure 2.1: Prototype symmetry motifs: geometric and discrete transformations that can commute with the Hamiltonian.

2.3 Degeneracy, symmetry, and when degeneracy is protected

For a bounded discrete spectrum, exact equality of two nearby energies is rare unless there is structure behind it. We call this *accidental* coincidence when no symmetry explanation exists. In contrast, symmetry-protected degeneracies are stable under small perturbations that preserve the same symmetry.

Proposition 2.1. *Let A and B be Hermitian symmetry generators commuting with H: $[A, H] = [B, H] = 0$. Define*

$$C := \frac{1}{i\hbar}[A, B].$$

Then C is Hermitian and $[C, H] = 0$.

Proof. Since A and B are Hermitian, $[A, B]^\dagger = -[A, B]$, so C is Hermitian. Then

$$[C, H] = \frac{1}{i\hbar}([AB, H] - [BA, H])$$

and expanding each commutator gives zero because both A and B commute with H. Hence $[C, H] = 0$. $\square$

Thus one can grow a closed set of symmetry generators by commutators. If that set closes to a Lie algebra with nontrivial structure constants, the representation theory is what drives degeneracy and spectral multiplets.

2.4 Particle on a circle: quantized L_z from single-valued wave functions

Let $\theta \in [0, 2\pi)$ label the position of a particle constrained to a circle. A small rotation by ϵ acts as

$$(\mathcal{R}(\epsilon)\psi)(\theta) = \psi(\theta - \epsilon).$$

Its infinitesimal form defines the generator L_z:

$$\mathcal{R}(\epsilon)\psi(\theta) \approx \left(1 - \frac{i\epsilon}{\hbar} L_z\right)\psi(\theta), \qquad L_z = -i\hbar\frac{d}{d\theta}.$$

Hence

$$L_z \left|\psi_m\right\rangle = m\hbar \left|\psi_m\right\rangle \quad \Longleftrightarrow \quad -i\hbar\frac{d\psi_m}{d\theta} = m\hbar\psi_m$$

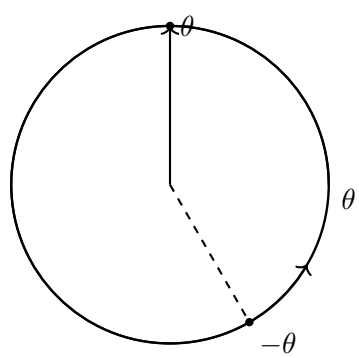
and therefore

$$\psi_m(\theta) = \frac{1}{\sqrt{2\pi}} e^{im\theta}.$$

Single-valuedness on $0 \sim 2\pi$ gives

$$\psi_m(\theta + 2\pi) = \psi_m(\theta) \iff e^{i2\pi m} = 1 \iff m \in \mathbb{Z}.$$

So the orbital angular-momentum quantum numbers are $m = 0, \pm 1, \pm 2, \ldots$.



Identification of θ and $\theta + 2\pi$: $\psi(\theta + 2\pi) = \psi(\theta)$

Figure 2.2: Circle coordinate, opposite angles, and periodicity condition.

2.5 Mirror symmetry and the $m \leftrightarrow -m$ pairing

Introduce a mirror (reflection) operator $\mathcal{M}$ with

$$(\mathcal{M}\psi)(\theta) = \psi(-\theta).$$

Its action on a momentum eigenmode is

$$\mathcal{M}\psi_m(\theta) = \psi_{-m}(\theta).$$

On a circle eigenstate:

$$\mathcal{M}L_z\psi_m = m\hbar\,\mathcal{M}\psi_m, \qquad L_z\mathcal{M}\psi_m = -m\hbar\,\mathcal{M}\psi_m,$$

so

$$[\mathcal{M}, L_z]\psi_m = 2m\hbar\,\mathcal{M}\psi_m \neq 0 \quad (m \neq 0).$$

Thus $\mathcal{M}$ does not commute with L_z, even though both may commute with H in a symmetric setup. When both symmetries are present (no mirror-breaking field), states m and $-m$ are paired at equal energy.

Important warning from the formalism: equal probabilities $|\psi_m(\theta)|^2 = |\psi_{-m}(\theta)|^2$ does not imply identical wave functions. Degeneracy concerns states and operators, not only densities.

A uniform magnetic field is a standard counterexample: it can split $\pm m$ levels while preserving some rotational structure of orbital motion in other settings. Symmetry arguments apply to what the Hamiltonian does, not to geometric intuition alone.

2.6 Generators, commutator algebra, and closure

For continuous symmetries a small transformation is written as

$$U(\epsilon) = 1 - \frac{i\epsilon}{\hbar}G + \mathcal{O}(\epsilon^2),$$

with Hermitian generator G. The commutator of two generators is another generator of another direction in the same symmetry structure if closure holds.

A useful worked pattern is:

$$[\,\mathcal{M}, L_z\,] \neq 0, \qquad \text{while } \; [H, L_z] = 0, \; [H, \mathcal{M}] = 0 \text{ (if mirror symmetric)}.$$

Hence $\mathcal{M}$ and L_z together generate two operations whose noncommutativity forces additional structure.

2.7 Three-dimensional angular momentum and ladder structure

For a particle in 3D,

$$L_x = yp_z - zp_y, \qquad L_y = zp_x - xp_z, \qquad L_z = xp_y - yp_x,$$

with canonical $[x_i, p_j] = i\hbar\delta_{ij}$. A compact expansion gives

$$[L_x, L_y] = i\hbar L_z,$$

and cyclic permutations. Equivalently,

$$[L_i, L_j] = i\hbar \sum_k \epsilon_{ijk} L_k.$$

For a rotationally invariant Hamiltonian, $[H, L_i] = 0$, so L_x, L_y, L_z are conserved quantities.

Define raising and lowering operators

$$L_\pm := L_x \pm iL_y.$$

Using the algebra above,

$$[L_z, L_\pm] = \pm\hbar L_\pm, \qquad [L_+, L_-] = 2\hbar L_z.$$

If $|\ell, m\rangle$ is an eigenvector of L_z,

$$L_z|\ell, m\rangle = m\hbar|\ell, m\rangle,$$

then

$$L_z(L_\pm|\ell, m\rangle) = (m \pm 1)\hbar(L_\pm|\ell, m\rangle),$$

so $L_\pm$ create a state with adjacent m-value when nonzero.

In a finite ladder representation, the chain terminates:

$$L_+|\ell, \ell\rangle = 0, \qquad L_-|\ell, -\ell\rangle = 0,$$

giving $m = -\ell, -\ell+1, \ldots, \ell$, hence $2\ell+1$ states in one rotational multiplet. For orbital wave functions on configuration space, $\ell, m \in \mathbb{Z}$ and $m \in \{-\ell, \ldots, \ell\}$. For more general angular-momentum types (including spin), half-integer m spectra are also possible; in this lecture the orbital circle argument selects the integer branch.

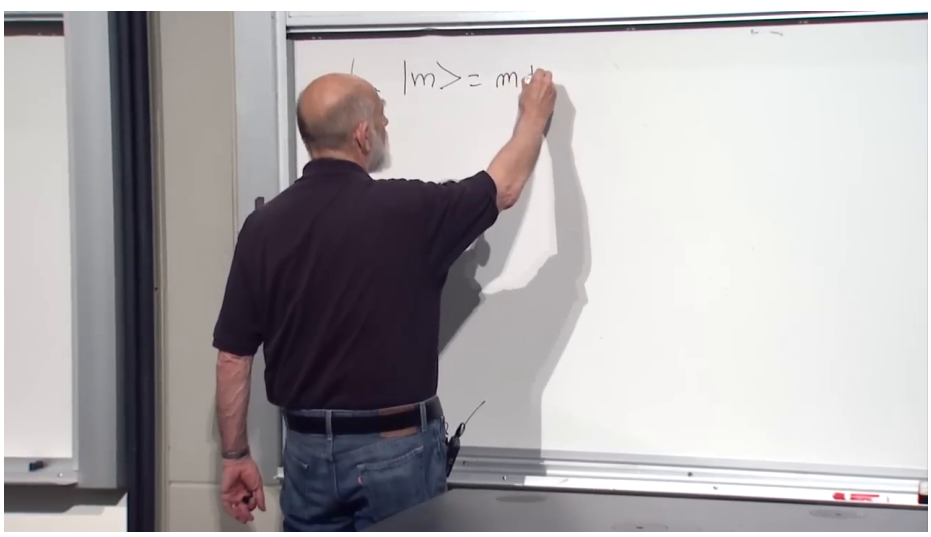


Figure 2.3: Illustrative ladder picture: $L_\pm$ connect adjacent m-levels inside a symmetry multiplet.

2.8 Summary

Symmetry enters as a commutation criterion with the Hamiltonian. Single symmetries can explain invariant spectral structure, but genuine multiplets require a noncommuting symmetry set whose commutator algebra closes. The circle provides the cleanest demonstration: rotational invariance gives $L_z = -i\hbar\partial_\theta$, single-valuedness gives integer m, and mirror symmetry links m to $-m$. In three dimensions, the algebra $[L_i, L_j] = i\hbar\epsilon_{ijk}L_k$ yields ladder operators $L_\pm$, a finite m-tower for each ℓ, and equal-energy multiplets in rotationally invariant systems.

Chapter 3

Angular Momentum, Central Forces, and the Oscillator Ladder

These notes follow Leonard Susskind's *Advanced Quantum Mechanics* Lecture 3, curated by LazyingArt LLC. The lecture opens as a continuation of the previous discussion of angular momentum, but almost immediately shifts from algebra to physics: angular momentum is the language of how a wavefunction depends on angles. From there the hour unfolds in two acts. First, angular-momentum multiplets are reconstructed and then used to reduce the central-force Schrödinger equation to a one-dimensional radial problem. Second, the same ladder-operator instinct reappears in the harmonic oscillator. Throughout, Susskind suppresses factors of $\hbar$ on the board; they are restored here.

3.1 Angular Momentum as Angular Dependence of the Wavefunction

Susskind begins by recalling the angular-momentum operators and the multiplets they form, but he does not stay in recap mode for long. Instead he asks what angular momentum is really about. His example is a particle moving in a central force field. Classically, the angular momentum

$$\mathbf{L} = \mathbf{r} \times \mathbf{p} \tag{3.1}$$

is conserved, so both its magnitude and its direction remain fixed. The orbit therefore stays in a fixed plane perpendicular to $\mathbf{L}$.

Quantum mechanically the particle is described by a wavefunction, and the point is no longer the orbit itself but the way the wavefunction changes under rotation. In a central-force problem the wavefunction may be written as a function of radius and angles,

$$\Psi = \Psi(r, \theta, \phi), \tag{3.2}$$

and angular momentum acts only on the angular part.

A two-dimensional prototype. Susskind briefly returns to two dimensions because there the basic idea is completely transparent. If motion is confined to a plane, the only relevant angular

momentum is the generator of rotations about the perpendicular axis:

$$L_z = -i\hbar \frac{\partial}{\partial \theta}. \tag{3.3}$$

An angular-momentum eigenstate therefore satisfies

$$L_z \psi(r,\theta) = \hbar \ell \, \psi(r,\theta), \tag{3.4}$$

or

$$-i\hbar \frac{\partial}{\partial \theta} \psi(r,\theta) = \hbar \ell \, \psi(r,\theta). \tag{3.5}$$

Here the derivative is only with respect to θ. The radial coordinate is a spectator. Solving for the θ-dependence gives

$$\psi(r,\theta) = \chi(r) e^{i\ell\theta}, \tag{3.6}$$

where ℓ must be an integer so that the wavefunction returns to itself when θ goes around the circle by 2π.

This is the first structural lesson of the lecture. Angular momentum is determined by the angular dependence of the wavefunction. The radial factor $\chi(r)$ may matter for the energy, but it does not matter for the angular-momentum eigenvalue.

Back to three dimensions. The same statement survives in three dimensions. One writes the wavefunction schematically as

$$\Psi(r,\theta,\phi) = R(r)\, Y_{\ell m}(\theta,\phi), \tag{3.7}$$

where the angular factor $Y_{\ell m}$ carries the rotational quantum numbers. These are the spherical harmonics. Susskind's emphasis is that one need not know their explicit formulas at this stage. What matters is that the angular problem and the radial problem separate, and angular momentum belongs entirely to the first of them.

With that physical point fixed, he returns to the operator algebra.

3.2 From the L_z Basis to Finite Multiplets

The next move, as Susskind says in a distinctly Dirac-like spirit, is to pick one component of angular momentum and work in its eigenbasis. Take L_z. The basic commutators are

$$[L_x, L_y] = i\hbar L_z, \qquad [L_y, L_z] = i\hbar L_x, \qquad [L_z, L_x] = i\hbar L_y. \tag{3.8}$$

Now define

$$L_\pm = L_x \pm i L_y. \tag{3.9}$$

A short computation gives

$$[L_z, L_\pm] = \pm\hbar L_\pm. \tag{3.10}$$

Therefore, if

$$L_z|\ell, m\rangle = \hbar m \, |\ell, m\rangle, \tag{3.11}$$

then

$$L_z\big(L_\pm|\ell,m\rangle\big) = \big(L_\pm L_z + [L_z, L_\pm]\big)|\ell,m\rangle \tag{3.12}$$

$$= \hbar(m \pm 1)\big(L_\pm|\ell,m\rangle\big). \tag{3.13}$$

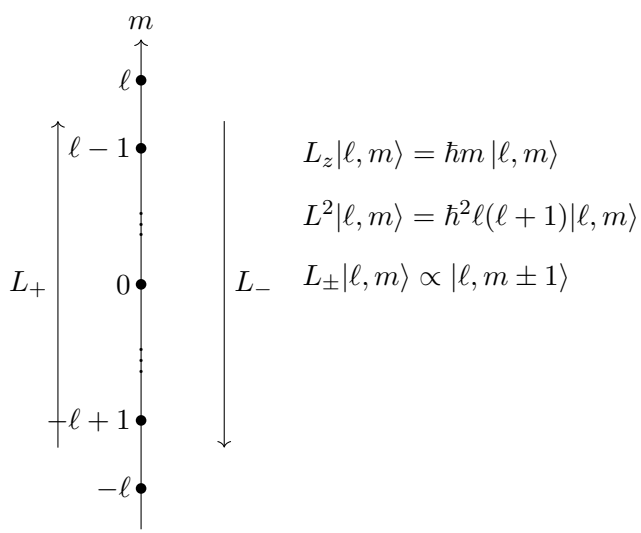


Figure 3.1: The finite angular-momentum ladder suggested by the blackboard discussion. The ladder operators move in unit steps of m, and the multiplet terminates symmetrically at $m = \pm\ell$.

So L_+ raises m by one unit and L_- lowers it by one unit:

$$L_\pm|\ell, m\rangle \propto |\ell, m \pm 1\rangle. \tag{3.14}$$

That is the first algebraic payoff. The second is the question Susskind immediately asks: does the ladder continue forever? It can only stop if eventually one of the raised or lowered states vanishes:

$$L_+|\text{top}\rangle = 0, \qquad L_-|\text{bottom}\rangle = 0. \tag{3.15}$$

Rotational symmetry then forces the spectrum to be symmetric about zero. There is nothing fundamentally privileged about the $+z$-direction relative to the $-z$-direction. Hence a finite ladder must run from some highest value $m = \ell$ down to the reflected value $m = -\ell$:

$$m = -\ell, -\ell + 1, \ldots, \ell - 1, \ell. \tag{3.16}$$

The number of states is therefore

$$2\ell + 1. \tag{3.17}$$

Susskind makes one further observation here. If the steps are always in integers, then a finite symmetric ladder can begin either at an integer or at a half-integer. Thus the abstract angular-momentum algebra allows both integer and half-integer multiplets. Later, when the discussion specializes to orbital wavefunctions on the sphere, only integer ℓ remain.

Remark 3.1. The lecture briefly mentions the possibility of infinite ladders. They are mathematically conceivable, but they are set aside as physically pathological for the finite-energy situations of interest here.

3.3 What a Multiplet Has in Common

Once the ladder has been built, Susskind asks what characterizes the entire multiplet. The answer is the square of the total angular momentum,

$$L^2 = L_x^2 + L_y^2 + L_z^2. \tag{3.18}$$

To evaluate it, he rewrites $L_x^2 + L_y^2$ in terms of the ladder operators. The ordered product is

$$\begin{aligned} L_- L_+ &= (L_x - iL_y)(L_x + iL_y) && (3.19)\\ &= L_x^2 + L_y^2 + iL_xL_y - iL_yL_x && (3.20)\\ &= L_x^2 + L_y^2 + i[L_x, L_y] && (3.21)\\ &= L_x^2 + L_y^2 - \hbar L_z. && (3.22) \end{aligned}$$

Therefore

$$L^2 = L_z^2 + \hbar L_z + L_- L_+. \tag{3.23}$$

There is also the equally valid alternative

$$L^2 = L_z^2 - \hbar L_z + L_+ L_-, \tag{3.24}$$

which corresponds to factoring in the opposite order. An audience member asks precisely this question in the lecture, and Susskind's answer is that the same derivation works if one applies the reordered identity to the bottom state instead of the top.

Worked derivation of $L^2 = \hbar^2 \ell(\ell+1)$. Apply the first ordered form to the highest state of the multiplet:

$$\begin{aligned} L^2|\ell,\ell\rangle &= \left(L_z^2 + \hbar L_z + L_- L_+\right)|\ell,\ell\rangle && (3.25)\\ &= \left(\hbar^2\ell^2 + \hbar^2\ell\right)|\ell,\ell\rangle + L_-(L_+|\ell,\ell\rangle). && (3.26) \end{aligned}$$

By definition of the top state,

$$L_+|\ell,\ell\rangle = 0, \tag{3.27}$$

and so

$$L^2|\ell,\ell\rangle = \hbar^2\ell(\ell+1)|\ell,\ell\rangle. \tag{3.28}$$

This is the blackboard formula $L^2 = \ell(\ell+1)$ with the suppressed factor of $\hbar^2$ restored.

The lecture then pushes one step further. This is not just a property of the top state. In fact,

$$[L^2, L_i] = 0, \qquad i = x, y, z, \tag{3.29}$$

and therefore L^2 commutes with $L_\pm$ as well. For example,

$$\begin{aligned} L^2(L_-|\ell,\ell\rangle) &= L_-(L^2|\ell,\ell\rangle) && (3.30)\\ &= \hbar^2\ell(\ell+1)(L_-|\ell,\ell\rangle). && (3.31) \end{aligned}$$

So the next state down has the same L^2-eigenvalue, and the same argument propagates all the way to the bottom of the ladder.

This is the meaning of the multiplet. All $2\ell+1$ states have the same total angular momentum, but different projections on the z-axis:

$$L^2|\ell,m\rangle = \hbar^2\ell(\ell+1)|\ell,m\rangle, \qquad L_z|\ell,m\rangle = \hbar m\,|\ell,m\rangle. \tag{3.32}$$

In the orbital case these states are represented by the spherical harmonics $Y_{\ell m}(\theta,\phi)$, with $\ell = 0, 1, 2, \ldots$ and $m = -\ell, \ldots, \ell$.

Susskind then makes the transition that matters for atomic physics. In a central-force problem the Hamiltonian commutes with angular momentum, so all members of a fixed ℓ-multiplet have the same energy:

$$[H, L_i] = 0 \qquad \text{for } V = V(r). \tag{3.33}$$

That closes the angular-momentum recap and opens the central-force problem.

3.4 From Three Dimensions to an Effective Radial Problem

At this point Susskind says explicitly that he will use a little classical intuition, even a tiny bit of cheating, to motivate the reduction of the central-force problem. The starting Hamiltonian is

$$H = \frac{p^2}{2m} + V(r), \tag{3.34}$$

with p^2 the full three-dimensional momentum squared and $V(r)$ depending only on the distance from the origin.

Classical reduction. Because the force is central, angular momentum is conserved. Classically one may therefore choose coordinates so that the orbit lies in the xy-plane. In that plane decompose the momentum into a radial component p_r and an angular component p_θ:

$$p^2 = p_r^2 + p_\theta^2. \tag{3.35}$$

The Hamiltonian becomes

$$H = \frac{p_r^2 + p_\theta^2}{2m} + V(r). \tag{3.36}$$

Now use the magnitude of the classical angular momentum:

$$L = r\, p_\theta. \tag{3.37}$$

Hence

$$L^2 = r^2 p_\theta^2, \qquad p_\theta^2 = \frac{L^2}{r^2}. \tag{3.38}$$

Substituting back gives the board equation

$$H = \frac{p_r^2}{2m} + \frac{L^2}{2mr^2} + V(r). \tag{3.39}$$

This is the effective radial Hamiltonian. Once the angular momentum is fixed, everything angular has been frozen into the parameter L, and the remaining motion is one-dimensional in r.

It is natural to define the effective potential

$$V_{\text{eff}}(r) = \frac{L^2}{2mr^2} + V(r). \tag{3.40}$$

The extra $1/r^2$ term is the centrifugal barrier. It is repulsive, not attractive: as r decreases, the potential rises sharply, so the corresponding force pushes the particle away from the origin.

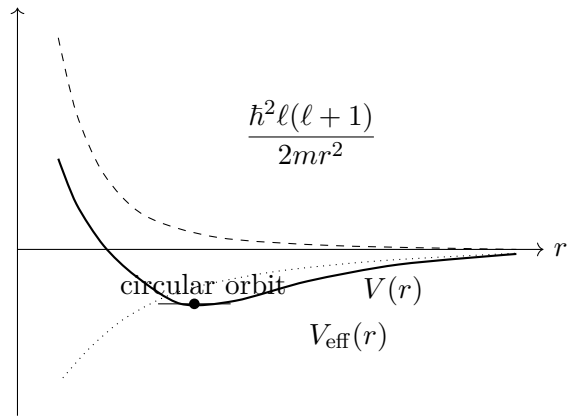


Figure 3.2: The effective radial potential for an attractive central force. The Coulomb tail $V(r) \sim -1/r$ and the centrifugal barrier $\sim 1/r^2$ combine into a one-dimensional potential with a minimum when $\ell \neq 0$.

What the effective potential means. For a Coulomb attraction $V(r) \sim -1/r$, the barrier behaves as $+1/r^2$. Near the origin the $1/r^2$ term dominates; far away the $1/r$ term dominates. Their sum has a minimum. A particle sitting at that minimum executes a circular orbit, while a particle with slightly different energy oscillates radially between a minimum and maximum radius. In the Coulomb case those radial oscillations are the classical elliptical orbits.

There is one exceptional case, emphasized later in the discussion: if $\ell = 0$, then the centrifugal barrier vanishes. Classically nothing prevents the motion from heading straight through the origin.

3.5 The Quantum Radial Equation

Susskind now turns to the quantum problem, but he keeps the classical reduction in view. He does not carry out every angular derivative on the board. Instead he uses the classical form of the Hamiltonian as a guide and then states the separated quantum equation. The omitted manipulations are real, but they do not change the basic story.

The full time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right)\Psi + V(r)\Psi = E\Psi. \tag{3.41}$$

Now choose a state of definite angular momentum:

$$\Psi_{\ell m}(r,\theta,\phi) = \psi(r)\, Y_{\ell m}(\theta,\phi). \tag{3.42}$$

Once ℓ and m are fixed, the angular dependence has already been solved by the spherical harmonic. The remaining unknown is the radial function $\psi(r)$.

The radial momentum operator is

$$p_r = -i\hbar\,\frac{d}{dr}, \tag{3.43}$$

so that

$$p_r^2 \to -\hbar^2\,\frac{d^2}{dr^2}. \tag{3.44}$$

Susskind is slightly casual about the sign when he speaks it aloud, but only p_r^2 appears in the Hamiltonian, so the final radial kinetic term is unambiguous.

On a state of definite ℓ, one replaces L^2 by $\hbar^2\ell(\ell+1)$. Thus the radial Hamiltonian acts as

$$H\psi(r) = E\psi(r), \tag{3.45}$$

with

$$H = -\frac{\hbar^2}{2m}\frac{d^2}{dr^2} + \frac{\hbar^2\ell(\ell+1)}{2mr^2} + V(r), \tag{3.46}$$

and therefore

$$-\frac{\hbar^2}{2m}\frac{d^2}{dr^2}\psi(r) + \frac{\hbar^2\ell(\ell+1)}{2mr^2}\psi(r) + V(r)\psi(r) = E\psi(r). \tag{3.47}$$

Remark 3.2. The lecture writes this as a pure second-order equation for $\psi(r)$. In standard separation of variables this clean one-dimensional form is the equation for the reduced radial function $u_\ell(r) = rR_\ell(r)$. To stay close to Susskind's notation while remaining precise, it is best to read $\psi(r)$ here as that reduced radial variable.

This is the main payoff of the angular-momentum recap. The original three-dimensional partial differential equation has become an ordinary differential equation in one variable. The angular structure has not disappeared; it has been compressed into the single parameter ℓ.

There is also a conceptual point that surfaces in the question period. Quantum mechanically one cannot literally rotate the whole wavefunction into a fixed plane, as one does classically for a single orbit. The memory of full three-dimensional motion survives in the fact that the relevant operator is L^2, not L_z^2. If one could replace L^2 by L_z^2, one would get something like ℓ^2. Instead one gets the genuinely three-dimensional result $\ell(\ell+1)$.

Susskind stops short of solving the radial equation explicitly. That, he says, is a matter of differential equations, and the lecture is after the structure of the spectrum rather than the explicit special functions.

3.6 Nodes, Degeneracies, and the Coulomb Accident

Having reduced the problem to one dimension, Susskind turns immediately to what one can say about the spectrum without solving the equation in detail.

Node counting. For fixed ℓ, the radial Schrödinger equation is a one-dimensional bound-state problem. The lowest bound state has no nodes, the first excited state has one node, the next has two, and so on. Here a node means a zero of the radial wavefunction. Susskind gives the physical reason in his usual way: more nodes mean more rapid wiggles; more rapid wiggles mean larger momentum; larger momentum means higher kinetic energy. In this sense the node count organizes the bound-state spectrum.

The generic central-force spectrum. Each value of ℓ gives a different radial equation because the centrifugal term depends on ℓ. Thus one obtains a separate tower of radial states for each ℓ. The $\ell = 0$ tower sits lowest because the centrifugal barrier is absent there. The $\ell = 1$ tower is shifted upward, the $\ell = 2$ tower upward again, and so on. At every level in the ℓ-th tower there are

$$2\ell + 1 \tag{3.48}$$

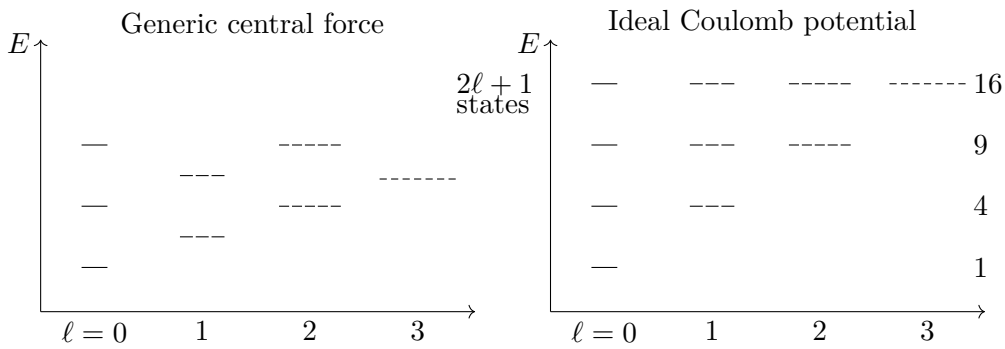


Figure 3.3: Left: the generic qualitative spectrum of an attractive central force. Each value of ℓ has its own radial tower, shifted upward by the centrifugal term, and each level carries degeneracy $2\ell + 1$. Right: in the ideal Coulomb problem, levels from different ℓ-towers align into shells with orbital degeneracies $1, 4, 9, 16, \ldots$.

degenerate states, corresponding to the allowed values of m.

At sufficiently high energy the states cease to be bound. Classically the particle escapes; quantum mechanically the discrete spectrum gives way to the continuum. For the bound states of an attractive potential, the energies lie below that escape threshold.

The special case $\ell = 0$. One audience question returns to the missing centrifugal barrier when $\ell = 0$. Classically, if the potential is Coulombic, the particle would simply run toward the center. Quantum mechanically that does not happen. The lowest state cannot be both sharply localized at the origin and at rest there. The uncertainty principle forces a compromise: the ground-state wavefunction spreads over a finite region, and the corresponding energy is finite rather than negatively infinite.

The Coulomb accident. For the special potential $V(r) \propto -1/r$, Susskind describes a remarkable extra degeneracy. The lowest $\ell = 1$ state has exactly the same energy as the first excited $\ell = 0$ state. The lowest $\ell = 2$ state has the same energy as the second excited $\ell = 0$ state. More generally, the lowest state in the ℓ-tower aligns with the ℓ-node state in the $\ell = 0$ tower. The result is a sequence of shells whose orbital degeneracies are

$$1, \qquad 4, \qquad 9, \qquad 16, \qquad \ldots \tag{3.49}$$

or, equivalently,

$$\sum_{k=0}^{n-1} (2k + 1) = n^2. \tag{3.50}$$

Susskind repeatedly warns that this is special to the ideal nonrelativistic Coulomb problem. A finite nuclear size, relativistic corrections, and spin all spoil it. The hydrogen atom approximately displays the pattern, but not exactly. In particular, because spin has not yet been included, the analysis is missing a factor of two. Real atomic shell counts begin

$$2, \qquad 8, \qquad 18, \qquad 32, \qquad \ldots \tag{3.51}$$

rather than

$$1, \qquad 4, \qquad 9, \qquad 16, \qquad \ldots \tag{3.52}$$

The lecture also pauses on a subtle question about wavefunctions. Bound states do not have compact support. They do not literally vanish beyond some finite radius. Instead they fall off rapidly, typically exponentially, so that the probability of finding the particle very far away is nonzero but extremely small.

That completes the atomic part of the lecture. Susskind then puts atoms aside for the moment and turns to a new system in which the same algebraic luck appears again.

3.7 The Harmonic Oscillator: Dirac's Trick Again

The transition to the harmonic oscillator is not a digression so much as a replay. Susskind emphasizes that many small oscillations in physics are approximately harmonic, and he even notes that oscillations near the bottom of an effective radial well are an example. The oscillator is the second case in which ladder operators solve the spectrum almost at once.

Take the mass to be 1 and write the spring constant as $k = \omega^2$. Then the classical Hamiltonian is

$$H = \frac{p^2}{2} + \frac{\omega^2 x^2}{2}. \tag{3.53}$$

Hamilton's equations are

$$\dot{x} = \frac{\partial H}{\partial p} = p, \qquad \dot{p} = -\frac{\partial H}{\partial x} = -\omega^2 x. \tag{3.54}$$

Therefore

$$\ddot{x} = -\omega^2 x, \tag{3.55}$$

which identifies ω as the classical frequency of oscillation.

Factorization and the zero-point term. Now comes the Dirac-like move. Since the Hamiltonian is a sum of squares, try to factor it:

$$\frac{1}{2}(p + i\omega x)(p - i\omega x). \tag{3.56}$$

Quantum mechanically this is not exactly H, because x and p do not commute. Using

$$[x, p] = i\hbar, \tag{3.57}$$

one finds

$$\frac{1}{2}(p + i\omega x)(p - i\omega x) = \frac{1}{2}\big(p^2 + \omega^2 x^2 + i\omega[x, p]\big) \tag{3.58}$$

$$= \frac{1}{2}\big(p^2 + \omega^2 x^2 - \hbar\omega\big). \tag{3.59}$$

Hence

$$H = \frac{1}{2}(p + i\omega x)(p - i\omega x) + \frac{1}{2}\hbar\omega. \tag{3.60}$$

The additive constant $\frac{1}{2}\hbar\omega$ is the zero-point energy. In the lecture Susskind temporarily drops it in order to focus on the ladder structure, and then restores it at the end.

Define

$$A_+ = \frac{p + i\omega x}{\sqrt{2\hbar\omega}}, \qquad A_- = \frac{p - i\omega x}{\sqrt{2\hbar\omega}}, \tag{3.61}$$

so that A_+ and A_- are Hermitian conjugates of one another. In standard notation they are $a^\dagger$ and a. The Hamiltonian becomes

$$H = \hbar\omega\left(A_+A_- + \frac{1}{2}\right). \tag{3.62}$$

Now define the number operator

$$N = A_+A_-. \tag{3.63}$$

At this point Susskind asks what Dirac would do. His answer is immediate: commute the factors. A direct calculation gives

$$[A_-, A_+] = 1. \tag{3.64}$$

From this it follows that

$$[N, A_+] = A_+, \qquad [N, A_-] = -A_-. \tag{3.65}$$

The ladder. Suppose

$$N|n\rangle = n|n\rangle. \tag{3.66}$$

Then

$$N(A_+|n\rangle) = (A_+N + [N, A_+])|n\rangle \tag{3.67}$$
$$= (n+1)(A_+|n\rangle), \tag{3.68}$$

so A_+ raises the eigenvalue by one. Similarly,

$$N(A_-|n\rangle) = (n-1)(A_-|n\rangle). \tag{3.69}$$

Thus the oscillator has precisely the same ladder structure that angular momentum had, except that the ladder now lives in energy rather than in m.

What prevents the ladder from descending forever? Here the answer is positivity:

$$\langle\psi|N|\psi\rangle = \langle\psi|A_+A_-|\psi\rangle = \|A_-|\psi\rangle\|^2 \geq 0. \tag{3.70}$$

So the eigenvalues of N are nonnegative. Since raising and lowering change them in integer steps, there must be a lowest one, and it must be

$$n = 0. \tag{3.71}$$

The corresponding ground state obeys

$$A_-|0\rangle = 0. \tag{3.72}$$

With the standard normalization,

$$A_+|n\rangle = \sqrt{n+1}\,|n+1\rangle, \qquad A_-|n\rangle = \sqrt{n}\,|n-1\rangle, \tag{3.73}$$

and so the full energy spectrum is

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right), \qquad n = 0, 1, 2, \ldots \tag{3.74}$$

This is exactly the spectrum Susskind reaches at the end of the lecture, with the $\frac{1}{2}\hbar\omega$ restored after having been temporarily set aside.

He also remarks, in response to a question, that one does not usually get ladder operators this cleanly. Physics has been lucky here twice: once for angular momentum, and once for the harmonic oscillator. The second luck is especially important because so many systems reduce, near equilibrium, to oscillator behavior.

3.8 Summary

Lecture 3 has a single mathematical instinct running through it. Angular momentum is first reinterpreted as angular dependence of a wavefunction, which leads naturally to the ladder of m-values and to the common multiplet value

$$L^2 = \hbar^2\ell(\ell + 1). \tag{3.75}$$

That algebra then reduces the central-force Schrödinger equation to a radial one-dimensional problem with an effective centrifugal barrier. The resulting spectra are organized by node number and by the $2\ell + 1$ degeneracy of each multiplet, with the Coulomb potential singled out by its special $1, 4, 9, 16, \ldots$ shell pattern before spin is included. The lecture closes by showing that the same algebraic way of thinking works again for the harmonic oscillator: factorize the Hamiltonian, compute a commutator, build a ladder, identify the lowest state, and read off the spectrum.

Chapter 4

Harmonic Oscillator Review, Spin-$\frac{1}{2}$, and the First Exchange Principle

These notes follow Leonard Susskind's fourth lecture in the supplementary *Advanced Quantum Mechanics* sequence, with curation by LazyingArt LLC. The lecture unfolds in a single arc: a purposeful review of the harmonic oscillator as preparation for later field theory, a rapid but concrete return to spin-$\frac{1}{2}$ as angular momentum, an atomic-physics motivation for spin and exclusion, and finally an intentionally unfinished opening of identical-particle statistics. Throughout the oscillator discussion Susskind sets $m = 1$ and mostly suppresses $\hbar$; we follow that convention and only mention $\hbar$ when the lecture does.

4.1 Why the Harmonic Oscillator Returns

Susskind begins by explaining why the harmonic oscillator keeps returning. Any system with an equilibrium configuration, once displaced slightly from equilibrium, will generally oscillate. A violin string is the easy example: it vibrates in a fundamental mode together with higher harmonics. But the real point is not acoustics. The oscillators that matter most for the rest of the course are the oscillatory modes of fields.

This is already visible in electromagnetism. The equilibrium configuration of the electromagnetic field is simply the vacuum: empty space with no electric or magnetic field. If one perturbs that equilibrium, for example by sending radiation into a cavity or waveguide, the field begins to oscillate in its allowed modes. Later those modes will be related to photons. The same pattern reappears in solids: a crystal does not consist of independent atoms oscillating separately, but of coupled atoms whose collective motions propagate as waves. Those oscillatory modes are again described by harmonic oscillators, and their quanta are phonons.

The review is therefore not a detour into familiar material. It is preparation for everything that follows: photons, phonons, and more generally the oscillatory structure that lies behind quantum field theory.

4.2 Oscillator Algebra in Ladder Form

With that motivation in place, Susskind returns to the algebraic description. In the convention $m = 1$, the Hamiltonian is

$$H = \frac{1}{2}p^2 + \frac{1}{2}\omega^2 x^2, \tag{4.1}$$

where ω is both the oscillator frequency and the parameter setting the curvature of the potential.

The creation and annihilation operators are defined by

$$a_\pm = \frac{p \pm i\omega x}{\sqrt{2\omega}}. \tag{4.2}$$

The factor $\sqrt{2\omega}$ is not decorative. It is chosen so that the commutator takes the simplest possible form,

$$[a_-, a_+] = 1. \tag{4.3}$$

The number operator is then

$$N = a_+ a_-. \tag{4.4}$$

This commutator, together with the definition of N, implies that a_+ and a_- raise and lower the eigenvalues of N by one unit. The spectrum of N is therefore

$$0, 1, 2, 3, \ldots$$

The lecture also recalls how the Hamiltonian is expressed in terms of N. Classically one would expect the product $a_+ a_-$ simply to reconstruct the Hamiltonian. Quantum mechanically the noncommutativity produces an extra term:

$$\begin{aligned} N &= \frac{1}{2\omega}(p + i\omega x)(p - i\omega x) && (4.5)\\ &= \frac{1}{2\omega}(p^2 + \omega^2 x^2 + i\omega[x, p]) && (4.6)\\ &= \frac{1}{2\omega}(p^2 + \omega^2 x^2 - \omega), && (4.7) \end{aligned}$$

since $[x, p] = i$ in the $\hbar = 1$ convention. Therefore

$$H = \omega\left(N + \frac{1}{2}\right). \tag{4.8}$$

Susskind also writes the simplified form

$$H = \omega N \tag{4.9}$$

when the additive constant is dropped. He is explicit about the distinction: the zero-point term $\frac{1}{2}\omega$ is real, but for many purposes it can be ignored because an additive constant in the energy does not change the equations of motion.

The extra $\frac{1}{2}\omega$ is the ground-state energy of the oscillator, or, in field-theory language, the vacuum energy. Susskind emphasizes that it is a consequence of the uncertainty principle: x and p cannot both vanish simultaneously, so the positive terms in the Hamiltonian cannot both be driven to zero.

4.3 A Student Question, Wave Functions, and the Ground State

At this point a student interrupts with a natural question: why does the ladder include 0 at all? Susskind first answers abstractly, before passing to wave functions. Because the Hamiltonian is built from $p^2 + \omega^2 x^2$, it cannot have arbitrarily negative eigenvalues. There must therefore be a bottom state. The only way the ladder can stop is if the lowering operator annihilates that state:

$$a_-|0\rangle = 0. \tag{4.10}$$

Applying $N = a_+ a_-$ to the same state gives

$$N|0\rangle = 0. \tag{4.11}$$

So the bottom state has number eigenvalue 0. Susskind pauses over the notation: the symbol 0 here is merely the name of the state, not the zero vector.

Only after that abstract argument does he place the wave-function description "parallel" to the operator structure. State vectors become wave functions $\psi(x)$, and energy eigenstates may be written as $\psi_0(x), \psi_1(x), \ldots$. In the position representation,

$$X \mapsto x, \qquad P \mapsto -i\,\frac{d}{dx}. \tag{4.12}$$

If one restores $\hbar$, this becomes $P \mapsto -i\hbar\, d/dx$; Susskind remarks that dimensional analysis tells one where the $\hbar$'s must go.

The time-independent Schrödinger equation is

$$\left[-\frac{1}{2}\frac{d^2}{dx^2} + \frac{1}{2}\omega^2 x^2\right]\psi(x) = E\,\psi(x). \tag{4.13}$$

As he stresses, this equation can be solved formally for any E, but most formal solutions are not physical. For generic energy the wavefunction explodes at infinity and cannot be normalized. The admissible states are square-integrable wavefunctions,

$$\int_{-\infty}^{\infty} \psi^*(x)\psi(x)\,dx = 1. \tag{4.14}$$

Only for special values of E does the differential equation yield such honest eigenfunctions.

The smart route to ψ_0. Rather than solve the second-order equation directly, Susskind uses the simpler first-order equation that must hold in the ground state:

$$a_-\psi_0(x) = 0. \tag{4.15}$$

Since an overall numerical factor is irrelevant in a homogeneous equation, one may write

$$a_- \propto p - i\omega x.$$

Substituting $p = -i\,d/dx$ gives

$$(p - i\omega x)\psi_0(x) = 0 \qquad \Longrightarrow \qquad \left(\frac{d}{dx} + \omega x\right)\psi_0(x) = 0. \tag{4.16}$$

This is much simpler than the full Schrödinger equation. The standard trick is to set

$$\psi_0(x) = e^{f(x)}. \tag{4.17}$$

Then

$$\psi_0'(x) = f'(x)e^{f(x)},$$

so the common factor $e^{f(x)}$ drops out and one is left with

$$\frac{df}{dx} + \omega x = 0. \tag{4.18}$$

Integrating once,

$$f(x) = -\frac{1}{2}\omega x^2 + C. \tag{4.19}$$

Hence the ground-state wavefunction is

$$\psi_0(x) \propto e^{-\omega x^2/2}. \tag{4.20}$$

This is the Gaussian Susskind draws on the board: smooth, concentrated near the origin, and rapidly decaying. It is manifestly square integrable. Its energy is already known from the ladder algebra,

$$E_0 = \frac{\omega}{2}. \tag{4.21}$$

If one wants the normalized form in the lecture convention, it is

$$\psi_0(x) = \left(\frac{\omega}{\pi}\right)^{1/4} e^{-\omega x^2/2}. \tag{4.22}$$

4.4 The First Excited State, Higher States, and the Semiclassical Picture

Once the ground state is explicit, Susskind immediately asks for the first excited state. He again uses the algebra rather than attacking the differential equation afresh:

$$\psi_1(x) \propto a_+\psi_0(x). \tag{4.23}$$

Since $a_+ \propto p + i\omega x$, one has

$$\psi_1(x) \propto (p + i\omega x)\psi_0(x). \tag{4.24}$$

But from the ground-state equation $(p - i\omega x)\psi_0 = 0$, it follows that

$$p\psi_0 = i\omega x\,\psi_0.$$

Therefore

$$(p + i\omega x)\psi_0 = 2i\omega x\,\psi_0, \tag{4.25}$$

and so, up to an irrelevant constant,

$$\psi_1(x) \propto x\,e^{-\omega x^2/2}. \tag{4.26}$$

This is enough to read off the physics. The first excited state is an odd function:

$$\psi_1(-x) = -\psi_1(x).$$

Its probability density is even, but the wavefunction itself changes sign at the origin, and in fact vanishes there. Thus the probability of finding the oscillator exactly at the equilibrium point is zero in the first excited state. The derived form also makes clear that the first excited state has one node, at $x = 0$.

The lecture does not derive a closed formula for all higher states, and it is better not to force one in at this stage. The essential qualitative structure is already visible. Repeated action of a_+ produces higher polynomials multiplying the same Gaussian envelope. Each higher state acquires another node; the parity alternates between even and odd; and the wavefunction is pushed farther outward from the origin. In standard notation these polynomials are Hermite polynomials, but the lecture's emphasis is not the name of the family. It is the picture.

That picture is simple and physical. More support at large $|x|$ means more potential energy. More rapid oscillation means more momentum. As the energy increases, the wavefunction develops more and more "wiggles" and extends farther and farther into the wings. Susskind's point is that the oscillator is trying to do what classical intuition says it should do: high-energy states are likely to be found farther from the center, but when they pass through the center they carry larger momentum.

The lecture then pauses over the correspondence principle. A single high-n eigenstate is not itself a classical oscillator moving back and forth. To see something that looks classical, one must form a wave packet by superposing many energy eigenstates. Susskind's example is a displaced Gaussian: start with the ground-state packet and move it away from the origin. It is no longer an energy eigenstate, so under the time-dependent Schrödinger equation its components pick up different phases. The packet then swings back and forth, becoming more oscillatory near the center and smoother near the turning points, while maintaining its overall character. In that way the harmonic oscillator does satisfy the correspondence principle, but the classical motion is seen in a packet, not in a stationary eigenfunction.

The lecture ends the oscillator discussion with one further remark. Exactly solvable systems are rare. The harmonic oscillator is one of them, and the idealized hydrogen atom is another; both will continue to serve as laboratories for more general ideas.

4.5 Spin-$\frac{1}{2}$ as Angular Momentum

Susskind then explicitly changes topic. The other subjects on his list for the evening are spin and the boson-fermion difference, and spin comes first. The return to spin is not meant as a fresh introduction. It is a reinterpretation of the earlier two-state system in the language of angular momentum.

For a particle at rest there is no orbital angular momentum $\mathbf{R} \times \mathbf{P}$, but there may still be spin. One may picture that spin literally as a tiny object spinning, or one may regard it as an abstract intrinsic property attached to the particle. Susskind is deliberately permissive about the picture. What matters is the mathematics: whatever this extra degree of freedom is, it transforms as angular momentum.

The decisive criterion is the commutation relations. If three operators are attached to the spatial directions x, y, and z, and if they satisfy the angular-momentum algebra, then they are angular

momentum. In particular,

$$[L_z, L_x] = iL_y, \tag{4.27}$$

with the other relations obtained by cyclic permutation.

The candidate operators are the Pauli matrices,

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \tag{4.28}$$

Susskind chooses the representation in which σ_z is diagonal. Its eigenvalues are $+1$ and -1, corresponding to spin up and spin down along the z-axis.

The board calculation proceeds by explicit multiplication:

$$\sigma_z\sigma_x = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = i\sigma_y, \tag{4.29}$$

$$\sigma_x\sigma_z = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = -i\sigma_y. \tag{4.30}$$

Therefore

$$[\sigma_z, \sigma_x] = 2i\sigma_y. \tag{4.31}$$

This is almost the angular-momentum algebra, but not quite. The mismatch is the factor of 2. The fix is exactly the one visible on the board:

$$S_i = \frac{\sigma_i}{2}. \tag{4.32}$$

Then the operators become

$$\frac{\sigma_z}{2} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}, \qquad \frac{\sigma_x}{2} = \begin{pmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix}, \qquad \frac{\sigma_y}{2} = \begin{pmatrix} 0 & -\frac{i}{2} \\ \frac{i}{2} & 0 \end{pmatrix}, \tag{4.33}$$

and now

$$[S_z, S_x] = iS_y, \tag{4.34}$$

again with cyclic permutations. The two-state system is therefore not merely analogous to angular momentum; it is the simplest nontrivial angular-momentum system.

Its z-component has eigenvalues

$$m_s = \pm\frac{1}{2}. \tag{4.35}$$

This is the spin-$\frac{1}{2}$ doublet. Susskind emphasizes that many of the particles one most naturally thinks of as fundamental — electrons, muons, neutrinos, quarks — carry precisely this kind of spin.

4.6 From Spin to Atoms and Shell Counting

Once the algebraic point is in place, the lecture turns to why nature should care. There are two kinds of angular momentum: orbital angular momentum and spin. Their vector sum is the total angular momentum,

$$\mathbf{J} = \mathbf{L} + \mathbf{S}. \tag{4.36}$$

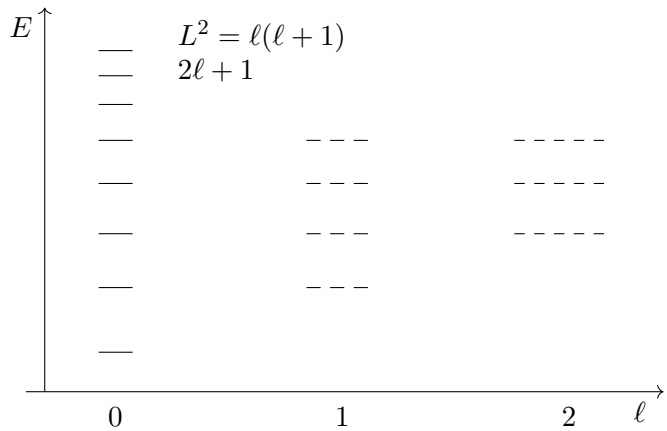


Figure 4.1: Schematic hydrogenic degeneracy pattern, redrawn from the lecture board and transcript. The vertical spacing is not to scale; the important content is the alignment of the $\ell = 1$ and $\ell = 2$ ladders with higher $\ell = 0$ states.

For a particle,

$$\mathbf{L} = \mathbf{R} \times \mathbf{P}.$$

As Susskind later stresses, angular momentum is always defined relative to a chosen origin. In atomic physics the natural origin is the nucleus. The detailed rules for adding $\mathbf{L}$ and $\mathbf{S}$ are postponed; the lecture only needs the notation and the physical distinction.

The atomic motivation begins with the hydrogen atom. In the lecture convention,

$$L^2 = \ell(\ell+1), \tag{4.37}$$

and for fixed ℓ the multiplicity is

$$2\ell + 1. \tag{4.38}$$

Susskind redraws the spectrum schematically as an energy-versus-ℓ plot. The actual hydrogen energies are negative, but for the drawing he simply shifts the picture upward and focuses on degeneracies rather than scale.

The $\ell = 0$ column contains the s-wave tower. The first $\ell = 1$ level is degenerate with the second $\ell = 0$ level, the first $\ell = 2$ level with the third $\ell = 0$ level, and so on. Counting the states shell by shell gives

$$1, \qquad 1+3=4, \qquad 1+3+5=9, \qquad 1+3+5+7=16, \tag{4.39}$$

or more compactly,

$$1, 4, 9, 16, \ldots = n^2. \tag{4.40}$$

This is only the orbital counting. Once spin is included, the idealized shell totals double to

$$2, 8, 18, 32, \ldots = 2n^2. \tag{4.41}$$

The lecture then backs up and turns that counting into a physical argument. Helium is well described, at first approximation, by putting two electrons into the lowest orbital state. But when one proceeds to lithium, the third electron does not drop into the same state. It enters the first excited shell. Why should that happen, when it would seem energetically cheaper to place all the electrons in the lowest orbital?

Pauli's answer is the exclusion principle: no two electrons can occupy the same state. But that raises an immediate problem, because helium plainly allows two electrons in the same orbital. The way out is to recognize that the electron has more structure than its orbital motion alone. It carries an additional two-valued property. In the lecture this is introduced in the simplest possible language: call the two values up and down. Then two electrons may occupy the same orbital state provided they have opposite spin. They are not, after all, in the same full quantum state.

This is the lecture's route to the second row of the periodic table. The first excited shell contains one $\ell = 0$ state and three $\ell = 1$ states, for a total of four spatial states. Doubling by the two spin values gives eight possibilities, matching the eight entries in the second row of the periodic table in this very idealized shell picture.

Susskind also mentions two further points that keep the discussion honest. First, spectroscopy pointed in the same direction as chemistry, although the periodic-table argument is more transparent. Second, once the new two-valued property is interpreted as angular momentum, it ought to reveal itself in a magnetic field. A charged particle carrying angular momentum behaves like a tiny magnet, so the two spin orientations should split in energy when a magnetic field is applied. That splitting is indeed observed.

At the same time, the lecture is explicit that this shell model is naive. It ignores electron-electron interactions and therefore breaks down rather quickly for larger atoms. The simple $2n^2$ pattern is useful here as motivation for spin and exclusion, not as a quantitatively complete many-electron theory.

4.7 Identical Particles and the First Exchange Principle

This brings Susskind to the second advertised subject: bosons and fermions. A student asks whether the exclusion principle is a separate postulate. Susskind's answer is historically layered. In nonrelativistic quantum mechanics it can be stated as a postulate. In relativistic quantum theory it becomes deeper. But for the moment he wants a cleaner quantum-mechanical formulation, one phrased directly in terms of identical particles.

He begins by stepping back to classical statistical mechanics. Suppose two particles of the same kind are placed in boxes. If one imagines that the particles secretly carry names — Harry and Sally is the lecture's example — then placing Harry in box 1 and Sally in box 2 seems different from placing Sally in box 1 and Harry in box 2. One may nevertheless choose to count those as the same configuration, and in classical statistical mechanics that choice leads to the familiar $1/n!$ factors. The point is that classically one can imagine faint labels, even if no experiment can detect them.

Quantum mechanics is stricter. One does not have the luxury of treating truly identical particles as secretly labeled. Swapping two electrons is an honest transformation on the quantum state, not a bookkeeping convention.

For one particle the wavefunction is $\psi(x)$. For two identical particles the state is described by a wavefunction of two arguments,

$$\psi(x_1, x_2), \tag{4.42}$$

where x_1 and x_2 label the two particle slots, not two spatial directions. The probability density $|\psi(x_1, x_2)|^2$ is the probability of finding one particle at x_1 and the other at x_2.

The next move is the key one. Consider the operation that swaps the two particles. Susskind

denotes it by S; to avoid confusion with the spin operator $\mathbf{S}$, we shall write it as P_{12}. Its action is

$$P_{12}\psi(x_1, x_2) = \psi(x_2, x_1). \tag{4.43}$$

A function of two variables need not be symmetric, so in general this produces a new wavefunction.

Now swap twice. Nothing mysterious happens: interchanging the two particle slots a second time returns the original state. Therefore

$$P_{12}^2 = 1. \tag{4.44}$$

Susskind also notes that exchange is a transformation on the Hilbert space and so must be unitary. A unitary operator whose square is the identity can only have eigenvalues

$$\lambda = \pm 1. \tag{4.45}$$

Thus the physically relevant two-particle states fall into two classes:

$$P_{12}\psi = \psi, \qquad \text{or} \qquad P_{12}\psi = -\psi. \tag{4.46}$$

That is exactly where the lecture stops. The wavefunction of two identical particles either returns to itself under exchange or returns with a minus sign. The full interpretation of those two possibilities as bosonic and fermionic statistics is deferred to the next lecture. Susskind is about to begin with the $+1$ case when he announces that he is fading and leaves the rest for next time.

4.8 Summary

The lecture begins with the ubiquity of the harmonic oscillator and with a clear warning that this review is preparation for later field theory, not repetition for its own sake. From there it rebuilds the oscillator algebra, explains the zero-point shift, and answers a student question about why the ladder bottoms out. The abstract argument for a lowest state is then matched by an explicit position-space construction of the Gaussian ground state and the odd first excited state.

Only after that oscillator machinery is in place does the lecture pivot to spin, showing that the Pauli matrices become angular momentum operators once normalized as $S_i = \sigma_i/2$. The atomic discussion then uses hydrogenic degeneracies and naive shell counting to motivate Pauli's extra two-valued electron property, later recognized as spin. Finally, exclusion is reformulated as the first statement of identical-particle exchange: for two identical particles, swapping the particle slots leaves the wavefunction either unchanged or multiplied by -1. That deliberately unfinished dichotomy is the chapter's endpoint and the next lecture's starting point.

Chapter 5

Spin, Exchange, and the Two Minus Signs

These notes follow Leonard Susskind's fifth lecture in the supplementary Advanced Quantum Mechanics sequence, with explicit credit to Leonard Susskind and curation by LazyingArt LLC. The lecture begins in an apparently old-fashioned way, with helium, lithium, and shell filling, because that is where spin and exclusion first show up together in actual matter. Only after that empirical reminder does Susskind pivot to the more abstract point: the Pauli principle is really a statement about many-particle wavefunctions, and its relation to half-integer spin is part of a larger story that he calls a tale of two minus signs.

5.1 Atomic-Shell Motivation for Spin and Exclusion

Susskind opens by recalling that the fermionic character of the electron seems to involve two facts at once. One is that the electron has spin 1/2, so that, forgetting orbital motion for the moment, it has two spin states. The other is the Pauli exclusion principle. In the language of atomic shells, one cannot keep putting electrons into the same state. But, as he stresses immediately, when one talks about a state, spin counts.

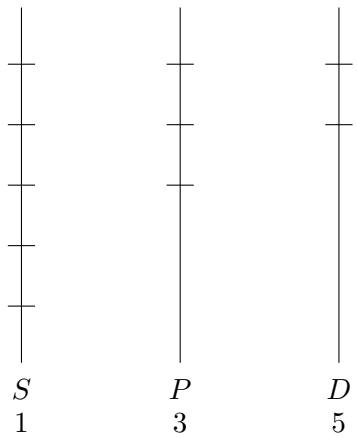
This is why two electrons may occupy the lowest spatial orbital of a helium atom. If spin were ignored, that would look forbidden. But the full one-particle state includes both the spatial wavefunction and the spin label, so one may place two electrons in the lowest orbital provided their spin labels differ. In the crude shell-model language, one is spin up and the other is spin down. Susskind even pauses to say that, more precisely, the two-electron spin state is entangled; the simple up-down language is only a mnemonic.

The lecture is careful about the level of approximation. This is the old shell-model picture, the "chemist's bad approximation," not a modern treatment of interacting electrons. It is being used only because it remembers the right counting. In that simple Coulomb picture, orbital angular momentum l contributes

$$2l + 1 \tag{5.1}$$

spatial states. Thus

$$l = 0 \Rightarrow 2l + 1 = 1, \qquad l = 1 \Rightarrow 2l + 1 = 3, \qquad l = 2 \Rightarrow 2l + 1 = 5. \tag{5.2}$$



These are the familiar S, P, and D sectors.

The visual point of the board sketch is only schematic: there are S-levels, P-levels, and then D-levels, with the Coulomb problem producing extra degeneracies across them. For the counting argument one needs only the first excited shell. It contains one $2s$ orbital and three $2p$ orbitals, so

$$N_{\text{first excited shell, spatial}} = 1_{2s} + 3_{2p} = 4. \tag{5.3}$$

Electrons carry two spin states, and therefore the same shell contains

$$N_{\text{first excited shell, electron}} = 2 \times 4 = 8 \tag{5.4}$$

electron states.

That is the shell-filling logic behind lithium, beryllium, and then neon. The outer electron in lithium must move into the next shell because there is simply no room left in the lowest one. The first excited shell can hold eight electrons because its four spatial states are doubled by spin. After neon the next electron must again go higher, which is why sodium looks chemically like lithium: each has one valence electron outside a closed shell.

Susskind then stops himself. He says, in effect, that he did not want to do chemistry that evening. He wanted only to remind the audience that two things seem to go together in nature: half-integer spin and exclusion. He sharpens the question by contrasting electrons with photons. Photons certainly do not obey exclusion; classical electromagnetic waves are precisely situations in which many, many quanta occupy the same state. In the bosonic case there is no occupancy bound at all. The lecture is therefore led to the real question: why do half-integer spin and exclusion come together, while integer spin and unlimited occupancy come together?

5.2 Many-Particle Wavefunctions and Identical Particles

With that empirical picture in place, Susskind announces the more abstract part of the lecture. The exclusion principle is not just a rule of atomic bookkeeping. It is a statement about the wavefunctions of many-particle systems.

He asks us, for the moment, to forget spin and restore it later. Let x_i denote the position of particle i. Here the subscript i labels the particle, not a Cartesian direction. The three spatial coordinates are being suppressed into a single symbol x.

For a system of n particles, a natural basis is

$$|x_1, x_2, \ldots, x_n\rangle, \tag{5.5}$$

the state in which particle 1 is at x_1, particle 2 is at x_2, and so on. If the abstract many-particle state is $|\Psi\rangle$, then its coordinate-space wavefunction is

$$\psi(x_1, x_2, \ldots, x_n) = \langle x_1, x_2, \ldots, x_n \mid \Psi\rangle. \tag{5.6}$$

The quantity

$$|\psi(x_1, x_2, \ldots, x_n)|^2 \tag{5.7}$$

is the joint probability density for finding particle 1 at x_1, particle 2 at x_2, and so forth.

The lecture board often uses uppercase Ψ for a many-particle wavefunction, and that notation is visible in the later exchange-symmetry sketch. In these notes, to keep the hierarchy clean, $|\Psi\rangle$ denotes the abstract state vector while $\psi(x_1, \ldots, x_n)$ denotes its coordinate representation.

Now comes the new point. Suppose the particles are identical. Then one must ask whether the configuration "particle 1 at x_1, particle 2 at x_2" is physically the same as "particle 1 at x_2, particle 2 at x_1." Susskind dramatizes the question by imagining a blue electron and a red electron, but then immediately retracts the colors: electrons have no such tags. They are indistinguishable. So one is not allowed to build the physics out of bookkeeping labels that nature does not supply.

At the level of state vectors, this already suggests that exchanging two identical particles should not produce a genuinely new physical state. But quantum mechanics admits one further subtlety: overall phase is unobservable. That opens the possibility that exchange may return the same physical state only up to a phase.

5.3 Exchange Phase, Bosons, and Fermions

The transition is sharp. Susskind says this lecture is a tale of two minus signs, and the first of them has to do with exchange. Let us take two identical particles. Since only relative phases matter, the most general possibility consistent with indistinguishability is

$$\psi(x_1, x_2) = e^{i\phi}\, \psi(x_2, x_1). \tag{5.8}$$

Here $e^{i\phi}$ is a fixed phase.

Now exchange the particles twice. After the first swap one has picked up $e^{i\phi}$; after the second swap one has picked up another factor of the same phase. But two swaps bring the configuration back to itself, so

$$\psi(x_1, x_2) = e^{i\phi}\, \psi(x_2, x_1) \tag{5.9}$$

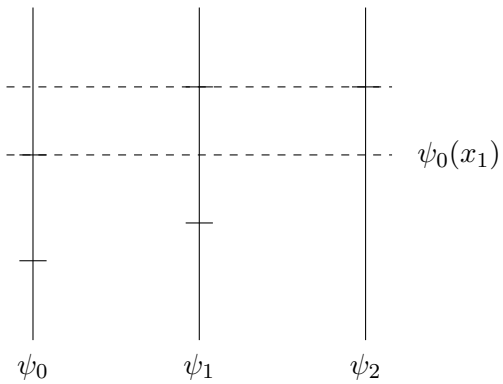
$$= e^{2i\phi}\, \psi(x_1, x_2). \tag{5.10}$$

Therefore

$$e^{2i\phi} = 1. \tag{5.11}$$

In the three-dimensional setting of this lecture, Susskind keeps the two familiar possibilities

$$e^{i\phi} = +1 \qquad \text{or} \qquad e^{i\phi} = -1. \tag{5.12}$$



Definition 5.1. Bosonic wavefunctions are symmetric under exchange:

$$\psi_B(x_1, x_2) = +\psi_B(x_2, x_1). \tag{5.13}$$

Fermionic wavefunctions are antisymmetric under exchange:

$$\psi_F(x_1, x_2) = -\psi_F(x_2, x_1). \tag{5.14}$$

This is the first minus sign of the lecture. A single swap, which one might naively have thought did nothing, changes the sign of a fermionic wavefunction. A double swap is trivial in both cases, as it must be.

Susskind is careful to locate the statement correctly. He explicitly remarks that the story changes in one and two spatial dimensions, so the reduction to ± 1 should be understood here as the three-dimensional story. In ordinary three-dimensional quantum mechanics, however, these are the two sensible possibilities, and they divide identical particles into bosons and fermions.

One further point matters for the lecture's rhythm. The sign itself is not yet observable by itself, because an overall minus sign does not change probabilities or expectation values. Nevertheless it is real in the quantum-mechanical sense that it changes how amplitudes combine. That is why Susskind says the exclusion principle has a deeper meaning than the sentence "you cannot put two particles into the same state." It is first a statement about symmetry of amplitudes.

5.4 Two-Particle Constructions and the Pauli Principle

Susskind now does what he always does when a formal rule threatens to become too abstract: he tests it on explicit states. The board picture at this stage shows a set of candidate one-particle orbitals, labeled ψ_0, ψ_1, ψ_2, with ψ_0 the lowest one. The point of the figure is not an accurate spectrum, but a clean bookkeeping device for discussing occupation.

A one-particle orbital becomes, in a two-particle wavefunction, a factor such as $\psi_0(x_1)$. With that understood, Susskind studies three cases.

Two particles in the same one-particle state. Start with

$$\psi_0(x_1)\psi_0(x_2). \tag{5.15}$$

Under exchange $x_1 \leftrightarrow x_2$, nothing changes:

$$\psi_0(x_2)\psi_0(x_1) = \psi_0(x_1)\psi_0(x_2). \tag{5.16}$$

The reason is elementary but important: wavefunctions are numerical functions, not operators, so the factors commute. This state is therefore symmetric. It is perfectly acceptable for bosons, but it cannot be a fermionic state.

Two particles in different one-particle states. Now try

$$\psi_0(x_1)\psi_1(x_2). \tag{5.17}$$

After exchange one gets

$$\psi_0(x_2)\psi_1(x_1). \tag{5.18}$$

In general the exchanged expression is neither equal to the original product nor equal to its negative. So the bare product is not, by itself, an acceptable bosonic or fermionic wavefunction. This does not mean that one cannot place one particle in ψ_0 and another in ψ_1. It means only that the correct two-particle state must respect the exchange rule.

The proper constructions are the symmetric and antisymmetric combinations:

$$\psi_B^{(01)}(x_1, x_2) = \frac{1}{\sqrt{2}}\Big[\psi_0(x_1)\psi_1(x_2) + \psi_0(x_2)\psi_1(x_1)\Big], \tag{5.19}$$

$$\psi_F^{(01)}(x_1, x_2) = \frac{1}{\sqrt{2}}\Big[\psi_0(x_1)\psi_1(x_2) - \psi_0(x_2)\psi_1(x_1)\Big]. \tag{5.20}$$

The first is unchanged by exchange and is therefore a bosonic state. The second changes sign and is therefore a fermionic state.

The special case that becomes exclusion. Now make the two one-particle states coincide. For bosons one obtains

$$\psi_B^{(00)}(x_1, x_2) \propto \psi_0(x_1)\psi_0(x_2) + \psi_0(x_2)\psi_0(x_1) = 2\,\psi_0(x_1)\psi_0(x_2), \tag{5.21}$$

which is nonzero and can be normalized in the usual way. Two bosons can occupy the same one-particle state.

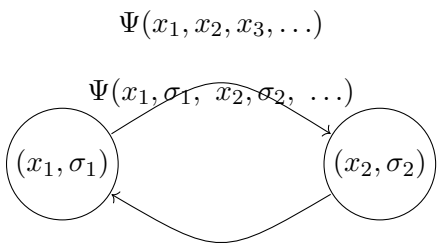
For fermions one instead obtains

$$\psi_F^{(00)}(x_1, x_2) \propto \psi_0(x_1)\psi_0(x_2) - \psi_0(x_2)\psi_0(x_1) = 0. \tag{5.22}$$

This is the lecture's direct mathematical statement of Pauli exclusion. The exclusion principle is not being appended as an extra axiom. It is the statement that the antisymmetric same-state amplitude cancels itself.

This is why Susskind says the exclusion principle can even be viewed as a kind of interference effect. The two exchanged contributions cancel destructively. He does not make that viewpoint the foundation, but he does accept it as a useful interpretation of why the amplitude vanishes.

The same logic extends immediately to many particles. A many-particle bosonic wavefunction is unchanged by any transposition of arguments, while a fermionic one changes sign under any transposition. Double exchange is always trivial; single exchange is not. That is the mathematical structure underlying the exclusion principle.



5.5 Including Spin, Permutations, and the Meaning of Exchange

Only after the spatial story is clear does Susskind restore spin. He asks us to make the smallest possible change: instead of a one-particle wavefunction depending only on position, let it depend on position and spin,

$$\psi(x, \sigma). \tag{5.23}$$

The discrete label σ may be taken, in the lecture, to mean the chosen σ_z value. One may think of $\sigma = \pm 1$, or equivalently $\uparrow, \downarrow$. The quantity

$$|\psi(x, \sigma)|^2 \tag{5.24}$$

is then the probability density to find the particle at x with the corresponding spin value.

For many particles the wavefunction becomes a function of the full labels,

$$\psi(x_1, \sigma_1;\, x_2, \sigma_2;\, \ldots). \tag{5.25}$$

The board picture at this point makes the pedagogical point visually: once spin is included, exchange means exchanging everything associated with particle 1 and particle 2, not merely the positions.

Thus the exchange rule becomes

$$\psi_B(x_1, \sigma_1;\, x_2, \sigma_2) = +\psi_B(x_2, \sigma_2;\, x_1, \sigma_1), \tag{5.26}$$

$$\psi_F(x_1, \sigma_1;\, x_2, \sigma_2) = -\psi_F(x_2, \sigma_2;\, x_1, \sigma_1). \tag{5.27}$$

What follows is exactly the statement with which the lecture began: two fermions cannot occupy the same full state, including spin. Two electrons may therefore share the same spatial orbital only when their spin labels differ.

A student remarks that this sounds like the permutation group, and Susskind agrees completely. Exchange is a permutation, and the ± 1 behavior furnishes a one-dimensional representation of the permutation group. In a compact operator notation one may write

$$P_{12}\psi_\pm = \pm\psi_\pm, \tag{5.28}$$

but the lecture's real point is the explicit one: exchange acts on the entire particle label.

Another student asks about the difference between adding wavefunctions and multiplying them. Susskind uses the question to separate two ideas that are often muddled. If one adds one-particle wavefunctions,

$$\alpha\,\psi_0 + \beta\,\psi_1, \tag{5.29}$$

one is describing a single particle in a superposition of two states. If instead one writes

$$\psi_0(x_1)\psi_1(x_2), \tag{5.30}$$

one is building a two-particle product state: particle 1 in one state, particle 2 in the other. Symmetrization or antisymmetrization comes only after that product has been formed.

The student discussion also foreshadows something Susskind will return to later. Because the same-state fermionic construction vanishes by cancellation of amplitudes, the Pauli principle can be viewed as a peculiar interference phenomenon. At this stage he leaves that remark hanging and turns to the second minus sign.

5.6 Rotation by 2π and the Second Minus Sign

The first minus sign concerned exchange of two identical fermions. The second concerns a single particle with half-integer angular momentum. Susskind asks what happens if one rotates a state by an angle θ, in particular by a full 2π.

At first sight one expects nothing at all. A 2π rotation is geometrically the identity rotation. But the same loophole that appeared in the exchange discussion appears here too: if all that must remain unchanged are probabilities and expectation values, the state may return only up to a phase.

Let $\psi(\theta)$ denote the state after a rotation by angle θ about the z-axis. The generator of such rotations is J_z. In Susskind's convention,

$$J_z\,\psi(\theta) = -i\,\frac{\partial}{\partial\theta}\psi(\theta). \tag{5.31}$$

Now suppose the state is a J_z-eigenstate with eigenvalue m:

$$J_z\,\psi = m\,\psi. \tag{5.32}$$

Combining the two equations gives

$$-i\,\frac{\partial}{\partial\theta}\psi(\theta) = m\,\psi(\theta), \tag{5.33}$$

whose solution is

$$\psi(\theta) = e^{im\theta}\psi(0). \tag{5.34}$$

Therefore

$$\psi(2\pi) = e^{i2\pi m}\psi(0). \tag{5.35}$$

At this point the lecture appeals to the familiar spectrum of angular momentum. The eigenvalues of J_z are either integers or half-integers. Hence

$$m \in \mathbb{Z} \quad\Rightarrow\quad e^{i2\pi m} = +1, \tag{5.36}$$

while

$$m \in \mathbb{Z} + \frac{1}{2} \quad\Rightarrow\quad e^{i2\pi m} = -1. \tag{5.37}$$

Integer-spin states therefore return to themselves under 2π, while half-integer-spin states return with a minus sign. A 4π rotation restores all of them:

$$\psi(4\pi) = \psi(0). \tag{5.38}$$

This is the second minus sign of the lecture. It has nothing directly to do with exchanging two particles. It belongs already to the rotational behavior of a single half-integer-spin state.

Remark 5.2. For notational cleanliness one may also write the rotation operator in the standard form

$$R_z(\theta) = e^{-i\theta J_z}. \tag{5.39}$$

The lecture itself proceeds instead through the differential relation $J_z = -i\,\partial/\partial\theta$ acting on $\psi(\theta)$. Because of this, one should not overemphasize the sign convention in the exponent. The physical invariant is the behavior under 2π: integer spin gives $+1$, half-integer spin gives -1, and 4π gives the identity.

Susskind immediately adds an important warning. The minus sign acquired under a 2π rotation is not itself visible in ordinary probabilities, any more than the exchange sign was. But that does not mean it is mathematically trivial. It is not the identity operation for fermions, only an operation whose effect disappears in $|\psi|^2$.

5.7 Topology, Spin-Statistics, and Observability

At this stage the lecture becomes more heuristic. Susskind shifts from algebra to pictures, demonstrations, and topology. He is not giving Pauli's rigorous proof of the spin-statistics theorem. He is trying to show why the two minus signs do not merely resemble one another accidentally.

The 4π topology. The cup trick and the belt trick are used to exhibit a basic fact about the topology of the rotation group. A path in rotation space corresponding to a 2π rotation is nontrivial; a path corresponding to 4π can be unwound. In Susskind's language, a 2π twist gives a tangle, while after another 2π the tangle can be undone. This is why the mathematics naturally distinguishes 2π from 4π.

The lesson is not yet spin-statistics. It is only that the half-integer behavior under 2π is not a cheap artifact of notation. It reflects the topology of the rotation group in three dimensions.

Exchange plus 2π rotation. The lecture then takes a more suggestive step. Susskind attributes to David Finkelstein a deeper topological observation: exchange of two particles and a 2π rotation of one particle are related. The belt demonstration makes this vivid. A belt configuration that can be read one way as an exchange can be read another way as a 2π twist, and "exchange plus a 2π rotation" becomes topologically trivial.

He then translates the picture into a worldline-type argument. Begin with no particles. Create two particle-antiparticle pairs, so that at an intermediate time there are four particles present. Now rotate one of them by 2π and simultaneously exchange two of the others. Topologically, Susskind argues, this history can be deformed into a trivial creation-annihilation process. A trivial process should carry no net phase. Therefore the phase acquired from exchange must cancel the phase acquired from rotation.

That is the lecture's heuristic version of the spin-statistics link. It is not presented as a proof, and Susskind explicitly says he has never seen it turned into a fully rigorous derivation in the form he likes. But it captures the right moral: the exchange sign and the 2π rotation sign are deeply connected, not merely parallel curiosities.

This is also why he broadens the discussion beyond elementary relativistic particles. Once the real criterion is the ability to create and annihilate particle-like excitations, the argument should

apply equally to solitons, solitary waves, monopoles, and other lump-like excitations in field theory or condensed matter systems. In that sense he takes the topology to be closer to the root of spin-statistics than the specific historical proof.

Why a global sign is invisible. Susskind then asks the operational question: can one observe the minus sign from a 2π rotation? If one rotates a single trapped electron adiabatically through 2π and then compares it with an otherwise identical experiment in which no such rotation was performed, the answer is no. The phase change is only global. Since probabilities depend on $\psi\psi^*$, the sign disappears.

This is the null experiment with which he begins the final part of the lecture. It is important because it prevents confusion. A universal 2π rotation of everything is not what one measures. The measurable effect appears only when one compares two coherent branches relative to one another.

Relative phase from branch interference. Now take one electron and split its wavefunction coherently into two branches. In the lecture these are called ψ_1 and ψ_2; to avoid conflict with the earlier one-particle orbitals, let us write them as ψ_{L} and ψ_{R}:

$$\psi(x) = \psi_{\mathrm{L}}(x) + \psi_{\mathrm{R}}(x). \tag{5.40}$$

This is still one electron. One branch of its wavefunction goes one way, the other branch another way. If one looks to determine which branch the electron took, the interference is destroyed; that is precisely the ordinary two-slit logic that Susskind repeatedly invokes.

Now let only one branch execute a 2π spin rotation, while the other branch is left alone. Then the state becomes

$$\psi(x) \longrightarrow \psi_{\mathrm{L}}(x) - \psi_{\mathrm{R}}(x). \tag{5.41}$$

This minus sign is no longer global. It is a relative phase between coherent branches, and therefore it changes the interference pattern when the branches are brought back together. Constructive interference can become destructive interference, and a point of reinforcement can become a node.

That is the core observability argument. A 2π rotation is unobservable as a global phase, but observable as a relative phase. Susskind then describes a more realistic version of the experiment: one sends a coherent electron beam through two arms, arranges a magnetic field in only one arm, and uses spin precession to make the spin in that arm execute one 2π rotation while the other arm does not. The pure adiabatic version and the practical precession-based version differ in apparatus, but not in conceptual content.

He closes this part with a subtlety. What is being detected is not a literal 2π rotation of the whole world. If the entire universe were rotated, no relative phase would be created and nothing observable would happen. The measurement works only because one branch of the world is rotated relative to another.

An exchange-interference coda. Right at the end, Susskind sketches a two-electron analog. One could imagine coherently splitting a pair of electrons into two branches, exchanging the electrons in one branch but not in the other, and then recombining the branches. The exchange sign would then appear as a relative phase, just as the 2π rotation sign did in the one-electron interferometer. He even suggests a hybrid experiment in which exchange occurs on one branch and a 2π rotation on the other, so that the two effects should cancel. The discussion remains exploratory, and he

does not claim that such an experiment has been carried out. But it reinforces the final conceptual point: quantum indistinguishability is subtler than classical sameness, and the exchange sign is a real ingredient of interference.

5.8 Summary

The lecture proceeds deliberately from chemistry to abstraction. It begins by recalling that, in the old shell picture, state includes spin, so two electrons can share the lowest spatial orbital only when their spin labels differ. That empirical reminder sets up the real subject: the exclusion principle is encoded in the symmetry of many-particle wavefunctions. Bosons have symmetric wavefunctions, fermions antisymmetric ones, and the Pauli principle appears because the antisymmetric same-state combination vanishes. After spin is restored as part of the particle label, the same rule says that two fermions cannot occupy the same full state, including spin. Only then does Susskind introduce the second minus sign, the -1 acquired by half-integer-spin states under a 2π rotation. The final topological and interferometric arguments are deliberately suggestive rather than formal, but they leave the intended impression: exchange and 2π rotation are deeply linked, and the sign picked up by a fermion under 2π becomes physically visible whenever it is converted from a global phase into a relative one.